# Chapter 5 — Tables and Forms

Tables and forms are the two parts of HTML where *structure* matters most. A table is not a grid of boxes — it is a set of relationships ("this 135 is North's revenue in Q2") that screen readers (software that reads the page aloud for blind users) announce cell by cell. A form is not a set of text boxes — it is a contract with a server (the remote program that receives the data) about which named values get sent, how, and when, plus a built-in validation engine (the browser's own checker that rejects bad input before it is sent) that most developers under-use.

Get these right and you get keyboard support, autofill (the browser filling in saved details such as your address), mobile keyboards, validation messages and accessibility almost for free. Get them wrong (a `<div>` "button", an input without a label, an error shown only in red) and no amount of JavaScript fully repairs it.

**What you'll learn**

1. Table semantics: `<caption>`, `<thead>`/`<tbody>`/`<tfoot>`, `<th scope>`, `colspan`/`rowspan`, `headers`, `<colgroup>`.
2. How a form works end to end: `action`, `method`, `name`, encoding, what gets sent (and what doesn't).
3. Labels, and every `<input>` type with its mobile keyboard.
4. `<select>`, `<datalist>`, `<textarea>`, `<fieldset>`/`<legend>`, `<button>`, `<output>`, `<progress>`, `<meter>`.
5. Built-in validation attributes, the `:user-invalid` pseudo-class (a CSS selector that matches an element in a particular state) and the Constraint Validation API (the JavaScript functions and properties for checking fields).
6. `autocomplete` and `inputmode`.
7. Accessible error messages: the pattern used by government and banking sites.

Every example is a file in `code/`. HTML files were screenshotted with headless Chrome (Chrome driven by a script, with no window; 900 CSS px wide); the Node scripts (`table_headers.mjs`, `form_data.mjs`, `constraint_api.mjs`) run under jsdom — a JavaScript implementation of the DOM — and print the outputs shown.

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## 5.1 Table semantics

### What it is

A **data table** is a grid where each **data cell** (`<td>`) is described by one or more **header cells** (`<th>`) — a column header, a row header, or both. The HTML elements exist so that software can compute that relationship.

Analogy: reading a timetable, your eyes go up to the column heading and left to the row heading to understand a number. A screen reader has no eyes; `<th>` + `scope` are the lines it follows.

### Why it matters

- A screen reader user moving into the cell "135" hears "Q2, North, 135". Without header markup they just hear "135".
- Proper structure lets the browser repeat `<thead>` on every printed page and lets CSS style header rows and columns with plain selectors (the part of a CSS rule that picks which elements it applies to, Chapter 6).

## How it works — the elements

| Element | Role |
|---|---|
| `<table>` | The table. |
| `<caption>` | The table's title — its accessible name (the label assistive technology announces). First child of `<table>`. |
| `<colgroup>` / `<col>` | Describe columns; only a few CSS properties apply to them (`background`, `border`, `width`, `visibility: collapse`). |
| `<thead>` | Header row(s). |
| `<tbody>` | Body rows. You may have several to group rows. Browsers insert one automatically if you omit it. |
| `<tfoot>` | Footer/summary rows (totals). |
| `<tr>` | A row. |
| `<th>` | Header cell. `scope="col"`, `"row"`, `"colgroup"` or `"rowgroup"` says which cells it describes. |
| `<td>` | Data cell. |
| `colspan` / `rowspan` | Make a cell span several columns/rows. |
| `headers="id1 id2"` | On a `<td>`: explicitly list the ids of its header cells (for irregular tables where `scope` isn't enough). |

Steps to build a correct table:

1. Start with `<caption>` stating what the table shows (include units).
2. Put the column headings in `<thead>` as `<th scope="col">`.
3. Make the first cell of each body row a `<th scope="row">` if it names the row.
4. Put totals in `<tfoot>`.
5. For two-level headings, use `colspan`/`rowspan` with `scope="colgroup"`/`"rowgroup"`. `colgroup` refers to a column group, so declare the groups with `<colgroup>` elements; `rowgroup` refers to the enclosing `<thead>`/`<tbody>`/`<tfoot>`. For truly irregular tables, use `id` + `headers`.
6. Style with CSS: `border-collapse: collapse`, right-aligned numbers, `font-variant-numeric: tabular-nums` (all digits the same width, so columns line up).

```html
<!-- code/01_table.html (first table) -->
<table>
  <caption>Quarterly revenue by region (USD thousands)</caption>
  <colgroup><col><col><col class="highlight"><col></colgroup>
  <thead>
    <tr><th scope="col">Region</th><th scope="col">Q1</th><th scope="col">Q2</th><th scope="col">Q3</th></tr>
  </thead>
  <tbody>
    <tr><th scope="row">North</th><td class="num">120</td><td class="num">135</td><td class="num">150</td></tr>
    <tr><th scope="row">South</th><td class="num">98</td><td class="num">102</td><td class="num">97</td></tr>
    <tr><th scope="row">West</th><td class="num">143</td><td class="num">160</td><td class="num">171</td></tr>
  </tbody>
  <tfoot>
    <tr><th scope="row">Total</th><td>361</td><td>397</td><td>418</td></tr>
  </tfoot>
</table>
```

```css
table { border-collapse: collapse; }
caption { caption-side: top; text-align: left; font-weight: 700; }
th, td { border: 1px solid #c8c8d8; padding: 6px 10px; }
tbody tr:nth-child(even) td { background: #f8f8fc; }        /* zebra stripes */
td.num, tfoot td { text-align: right; font-variant-numeric: tabular-nums; }
col.highlight { background: #fff6d6; }                     /* whole Q2 column */
```

The second table in the same file has two header levels:

```
<colgroup span="2"></colgroup>   <!-- Term + Course -->
<colgroup span="2"></colgroup>   <!-- the column group that "Students" spans -->
<thead>
  <tr><th rowspan="2" scope="col">Term</th><th rowspan="2" scope="col">Course</th>
      <th colspan="2" scope="colgroup">Students</th></tr>
  <tr><th scope="col">Enrolled</th><th scope="col">Completed</th></tr>
</thead>
<tbody>
  <tr><th rowspan="2" scope="rowgroup">Spring</th><th scope="row">HTML</th><td>40</td><td>35</td></tr>
  <tr><th scope="row">CSS</th><td>32</td><td>30</td></tr>
</tbody>
<tbody> … Autumn … </tbody>
```

**Quarterly revenue by region (USD thousands)**

| Region | Q1 | Q2 | Q3 |
|---|---|---|---|
| North | 120 | 135 | 150 |
| South | 98 | 102 | 97 |
| West | 143 | 160 | 171 |
| **Total** | **361** | **397** | **418** |

**Course enrolment** (two-level headers)

| Term | Course | Students | |
|---|---|---|---|
| | | Enrolled | Completed |
| Spring | HTML | 40 | 35 |
| | CSS | 32 | 30 |
| Autumn | HTML | 52 | 41 |
| | CSS | 45 | 44 |

*Figure 5.1 — Two data tables. The highlighted column comes from a single `<col class="highlight">`.*

### What assistive technology computes

`table_headers.mjs` rebuilds each table's grid (accounting for `rowspan`/`colspan`) and, for every data cell, collects the `<th>` cells whose `scope` points at it — a simplified form of the HTML standard's "assigning header cells" algorithm (a precise step-by-step procedure):

```
// code/table_headers.mjs (key part)
function headersFor(g, r, c) {          // g = grid of cells, r/c = slot of a data cell
  const found = [];
  const add = h => { if (!found.includes(h)) found.push(h); };
  for (let rr = 0; rr < r; rr++) {      // look up the column
    const h = g[rr][c];
    if (h.tagName === 'TH' && ['col', 'colgroup'].includes(h.scope)) add(h);
  }
  for (let cc = 0; cc < c; cc++) {      // look left along the row
    const h = g[r][cc];
    if (h.tagName === 'TH' && ['row', 'rowgroup'].includes(h.scope)) add(h);
  }
  return found.map(h => h.textContent.trim());
}
```

Output (abridged — the first table prints all nine cells the same way):

```
Table: Quarterly revenue by region (USD thousands)
   120  <- Q1 / North
   135  <- Q2 / North
   150  <- Q3 / North
    98  <- Q1 / South
   ...
Table: Course enrolment (two-level headers)
    40  <- Students / Enrolled / Spring / HTML
    35  <- Students / Completed / Spring / HTML
    32  <- Students / Enrolled / Spring / CSS
    30  <- Students / Completed / Spring / CSS
    52  <- Students / Enrolled / Autumn / HTML
    41  <- Students / Completed / Autumn / HTML
    45  <- Students / Enrolled / Autumn / CSS
    44  <- Students / Completed / Autumn / CSS
```

That is roughly the context a screen reader speaks when you move into each cell.

### Responsive tables

Wide tables overflow small screens. The safe fix keeps the table a table and scrolls it inside a labelled, focusable region (so keyboard users can scroll it):

```
<div role="region" aria-labelledby="cap1" tabindex="0" style="overflow-x: auto">
  <table><caption id="cap1">…</caption>…</table>
</div>
```

Changing `display` on the table elements themselves (for example `display: block` on `<table>` to stack cells) has historically made some browsers drop the table semantics — test with a screen reader if you do it.

## When not to use it

**Never for page layout.** Layout tables were common before CSS; today use CSS Grid or Flexbox (the two modern CSS layout systems) (Chapter 8, Chapter 9). If you inherit one, `role="presentation"` on the `<table>` tells assistive technology (screen readers and similar tools) to ignore its table semantics (its meaning as a table).

## Common mistakes

- Using `<td><b>…</b></td>` for headers instead of `<th>`.
- No `<caption>` (or a heading outside the table that isn't connected to it).
- The obsolete `summary` attribute — put explanations in the caption or nearby text.
- Empty cells used for spacing; merged cells everywhere "because it looks nicer" — each merge makes the structure harder to navigate.
- Using `<table>` for a list of cards or a form layout.

## In the interview

**30-second answer:** "Tables are for tabular data only. I give them a `<caption>`, use `<thead>/<tbody>/<tfoot>`, mark headers with `<th>` and `scope` (`col` / `row`, or `colgroup` / `rowgroup` for spanning headers), and fall back to `id` / `headers` for irregular tables. That lets screen readers announce the right headers per cell. For small screens I wrap the table in a scrollable, focusable region rather than changing its display."

- *Q: What does `scope` do?* A: Tells assistive technology which cells a header applies to — the rest of its column (`col`), row (`row`), or the group it spans.
- *Q: When do you need `headers`?* A: When cells have headers that aren't in the same row/column in a regular way, e.g. multi-level irregular tables.
- *Q: `border-collapse: collapse` vs `separate`?* A: Collapse merges adjacent cell borders into one line; separate (the default) draws each cell's border with `border-spacing` between them.

## Self-check

1. Where does `<caption>` go? → First child of `<table>`.
2. A header spanning two columns — which `scope`? → `colgroup`.
3. How do you right-align numbers so digits line up? → `text-align: right` + `font-variant-numeric: tabular-nums`.
4. How do you tell a screen reader a legacy layout table is not data? → `role="presentation"`.

### TL;DR

`caption` + `thead`/`tbody`/`tfoot` + `th scope`; `headers` for irregular tables; never for layout; scroll wide tables in a focusable region.

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## 5.2 How a form works — `action`, `method`, `name`

### What it is

A `<form>` collects user input and **submits** it: it builds a list of name/value pairs from its controls (the interactive fields and buttons) and sends it as an HTTP request (a message from browser to server; see Chapter 1). Controls are `<input>`, `<select>`, `<textarea>`, `<button>` (and `<output>`, which is never submitted).

### How it works — submission step by step

1. The user presses a submit button, or presses **Enter** in a text field (called *implicit submission*).
2. The browser runs **constraint validation** (5.5). If any control is invalid, it stops, shows a message and focuses the first invalid control.
3. A `submit` event (a notification that scripts can listen for) fires on the form. JavaScript can call `event.preventDefault()` (cancel the browser's normal action) to handle it itself (e.g. with `fetch`, JavaScript's function for sending HTTP requests).
4. The browser builds the **form data set** from *successful controls*:
   - the control has a `name` (no name → not sent);
   - it is not **disabled** (disabled → not sent; `readonly` → sent);
   - checkboxes and radios only if **checked** (value defaults to `"on"` if you omit `value`);
   - `<select multiple>` sends one pair per selected option;
   - only the submit button actually used is included (if it has a name).
5. It encodes the data (converts it into a text format safe to send — e.g. a space becomes `+` and `@` becomes `%40`) using `enctype` and sends it to `action` using `method`:
   - `method="get"` (default): data goes in the URL query string: `/signup?name=Ada+Lovelace&email=ada%40example.com`. Bookmarkable and cacheable (the browser or a server in between may store and reuse the response) — right for searches and filters, wrong for passwords or anything that changes data.
   - `method="post"`: data goes in the request body (the part of the HTTP message after the headers, not shown in the URL). Use for anything that creates or changes something, or is private.
   - `method="dialog"`: inside a `<dialog>`, closes it instead of sending anything.
6. `enctype` (POST only): `application/x-www-form-urlencoded` (default, like a query string), `multipart/form-data` (**required for file uploads**), or `text/plain` (debugging only).
7. The browser navigates to the server's response (unless JavaScript prevented it).

Other attributes: `action` defaults to the current URL; `target` works like a link's target; `novalidate` turns off step 2's built-in UI; a button can override the form with `formaction`, `formmethod`, `formenctype`, `formnovalidate`, `formtarget`. A control *outside* the form can still belong to it with `form="formId"`.

#### Labels

Every control needs a **label** — the text that names it. A `<label>` is connected in one of two ways:

```
<label for="name">Full name</label>          <!-- explicit: for = the input's id -->
<input id="name" name="name">

<label>Email <input type="email" name="email"></label>   <!-- implicit: wrap the input -->
```

What a real `<label>` gives you: the control's accessible name (screen readers read it on focus — when the control becomes the one receiving keyboard input), a bigger click target (clicking the text focuses the input or toggles the checkbox), and voice-control users can say "click Email". A `placeholder` is **not** a label: it disappears when typing starts, usually has poor contrast (too little difference between text and background colour), and isn't reliably announced.

### Buttons

`<button>` has three types, and **the default inside a form is `submit`**:

| type | Does |
|---|---|
| `submit` (default) | Submits the form. |
| `reset` | Resets every control to its initial value (rarely wanted — users click it by accident). |
| `button` | Nothing by itself — for JavaScript actions. |

Forgetting `type="button"` on a "Show password" or "Add row" button inside a form is a classic bug: clicking it submits the form.

```html
<!-- code/02_form_basics.html (excerpt) -->
<form action="02_form_basics.html" method="get">
  <label for="name">Full name</label>
  <input id="name" name="name" autocomplete="name" value="Ada Lovelace">
  <label>Email <input type="email" name="email" autocomplete="email" value="ada@example.com"></label>
  <label><input type="checkbox" name="newsletter" value="yes" checked> Send me the newsletter</label>
  <input type="hidden" name="source" value="chapter5">
  <button type="submit">Sign up</button>
  <button type="reset">Reset</button>
  <button type="button" id="preview">Preview data</button>
</form>
<script>
  const form = document.querySelector('form');
  const show = () => result.textContent = '?' + new URLSearchParams(new FormData(form));
  show();
  preview.addEventListener('click', show);
</script>
```

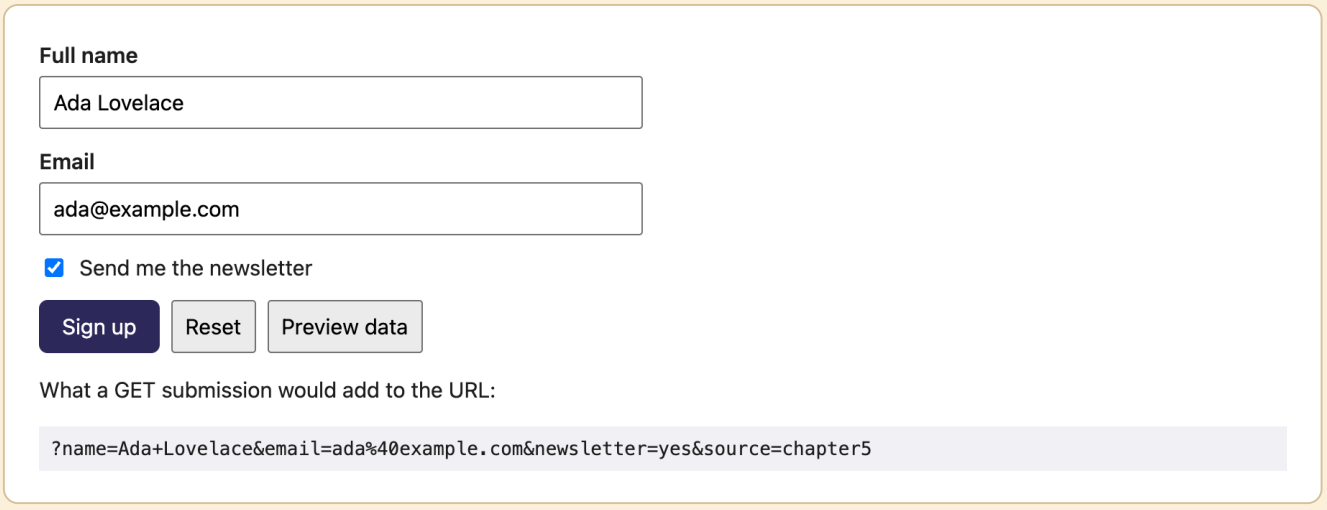



*Figure 5.2 — `FormData` + `URLSearchParams` produce exactly the query string a GET submission would send. Note the space encoded as `+` and `@` as `%40`.*

### Exactly what gets sent

`form_data.mjs` puts every tricky case in one form:

```
// code/form_data.mjs (form markup, abridged)
<input name="name" value="Ada Lovelace">
<input name="email" type="email" value="ada@example.com">
<input name="nick">                                  <!-- empty text: still sent, as "" -->
<input type="checkbox" name="newsletter" value="yes" checked>
<input type="checkbox" name="terms" value="yes">       <!-- unchecked: NOT sent -->
<input type="radio" name="plan" value="free">
<input type="radio" name="plan" value="pro" checked>
<select name="skills" multiple> <option selected>HTML</option><option selected>CSS</option><option>JS</option> </select>
<input name="coupon" value="SAVE10" disabled>          <!-- disabled: NOT sent -->
<input name="country" value="IN" readonly>             <!-- readonly: sent -->
<input value="no name attribute">                      <!-- no name: NOT sent -->
<input type="hidden" name="source" value="ch5">
<textarea name="bio">Hi &amp;amp; bye</textarea>

const data = new FormData(form);
for (const [k, v] of data) console.log(`  ${k.padEnd(10)} = ${JSON.stringify(v)}`);
console.log('\ndata.getAll("skills") =', data.getAll('skills'));
console.log('\nGET URL:\n  ' + form.action + '?' + new URLSearchParams(data));
```

Output:

```
FormData entries:
  name       = "Ada Lovelace"
  email      = "ada@example.com"
  nick       = ""
  newsletter = "yes"
  plan       = "pro"
  skills     = "HTML"
  skills     = "CSS"
  country    = "IN"
  source     = "ch5"
  bio        = "Hi & bye"

data.getAll("skills") = [ 'HTML', 'CSS' ]

GET URL:
  http://localhost/signup?
name=Ada+Lovelace&email=ada%40example.com&nick=&newsletter=yes&plan=pro&skills=HTML&skills=CSS&country=IN&source=ch5&bio=Hi+%26+bye
```

(`http://localhost` is jsdom's pretend page address.) The unchecked checkbox, disabled field and unnamed input are simply absent. The server-side consequence: **"checkbox missing" means "unchecked"** — code that expects `terms=no` will never see it.

### Security notes

- Client-side (running in the user's browser) anything (validation, `readonly`, hidden fields, `disabled`) is **not security** — anyone can edit the request. Server-side code runs on your own machine, which users cannot tamper with. The server must validate everything again.
- Never send passwords or personal data with GET — the URL lands in history, logs (servers' records of every request) and `Referer` headers (which tell the next site what page you came from).
- POST forms that change data need CSRF protection on the server (CSRF, cross-site request forgery, is when another site secretly makes your browser submit a form to a site you are logged in to; defences are secret per-form tokens or `SameSite` cookies, which the browser will not send on requests started by other sites; see the JavaScript book).

## In the interview

**30-second answer:** "A form submits its *successful controls* — named, enabled, and for checkboxes/radios, checked — to `action` using `method`. GET puts data in the URL (searches, filters); POST puts it in the body (changes, private data); file uploads need `multipart/form-data`. Every control needs a `<label>`, and buttons default to `type=submit`, so non-submit buttons need `type=button`. Server-side validation is mandatory."

- *Q: Disabled vs readonly?* A: Disabled: not focusable, not submitted, not validated. Readonly: focusable, submitted, not editable, not validated.
- *Q: How do you submit via JS without reloading?* A: Listen for `submit`, `preventDefault()`, send `new FormData(form)` with `fetch`. Use `form.requestSubmit()` (not `form.submit()`) to trigger submission from code — it

runs validation and fires the `submit` event; `submit()` skips both.

### Self-check

1. An `<input>` without `name` — is it sent? → No.
2. Which `enctype` for uploads? → `multipart/form-data` (with `method="post"`).
3. Why is a `<button>` inside a form submitting unexpectedly? → Its default type is `submit`.
4. Placeholder as label — why not? → It disappears on typing, often low contrast, and isn't a reliable accessible name.

### TL;DR

Named + enabled + checked → sent. GET for reading, POST for changing, multipart for files. Real `<label>`s, `type="button"` for script buttons, validate on the server.

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## 5.3 Input types

### What it is

`<input type="…">` changes the control's UI, its validation and — on phones — the on-screen keyboard. Unknown types fall back to `text`, so using a specific type is never worse than `text`.

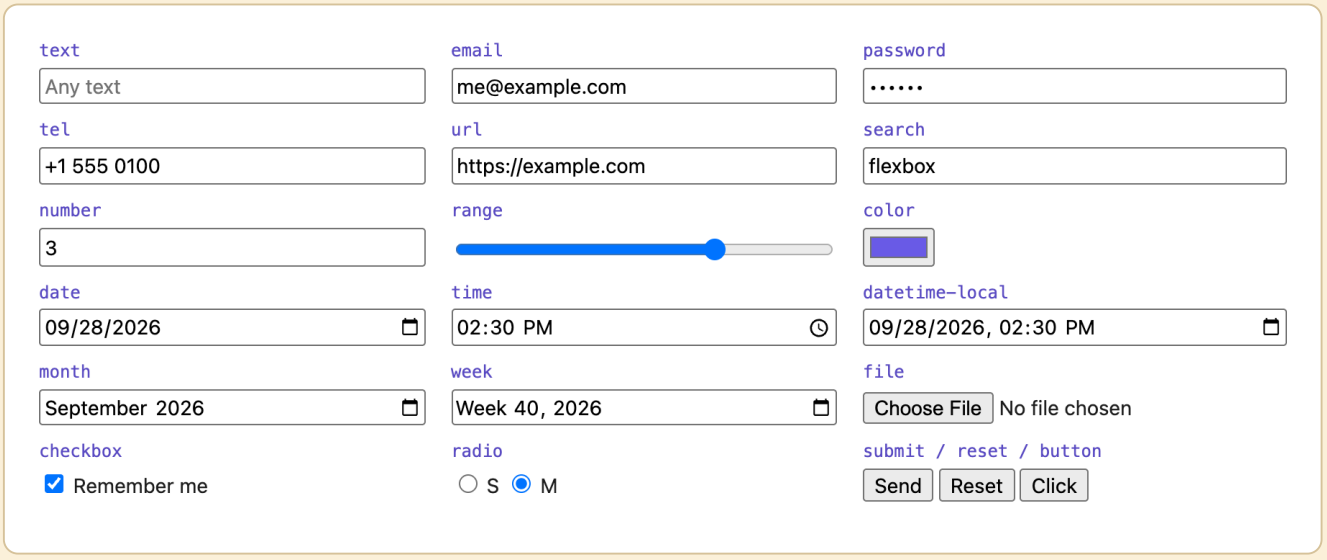



*Figure 5.3 — How Chrome on macOS draws each type (`code/03_input_types.html`). Other browsers draw them differently.*

| Type | Use for | Built-in validation | Mobile keyboard / UI |
|---|---|---|---|
| `text` | any single-line text | — | normal |
| `email` | email address | `typeMismatch` if not `x@y` shaped; `multiple` allows a comma list | `@` key visible |
| `password` | secrets | — | masked (shown as dots); password managers (apps that store and fill passwords) engage |
| `tel` | phone numbers | **none** (formats vary worldwide) | phone keypad |
| `url` | web addresses | `typeMismatch` without a scheme like `https:` | `/` and `.com` keys |
| `search` | search boxes | — | "search" enter key; may show a clear button |
| `number` | real quantities (count, age) | `min`, `max`, `step`, `badInput` | numeric keypad, spin buttons (small up/down arrows) |
| `range` | imprecise values (volume) | always has a value within range | slider |
| `color` | colour pick | value always `#rrggbb` | colour picker |
| `date`, `time`, `datetime-local` | dates/times | `min`, `max`, `step` | native pickers; value format `YYYY-MM-DD`, `HH:MM`, `YYYY-MM-DDTHH:MM` |
| `month`, `week` | months/weeks | same | **not supported in desktop Firefox and Safari** — they show a text box |
| `checkbox` | independent yes/no | `required` = must be checked | — |
| `radio` | one choice from a group (same `name`) | `required` on any one = must pick one | arrow keys move within the group |
| `file` | uploads | `accept` filters the picker (not enforced); `multiple` | camera/photos on phones (`capture`) |
| `hidden` | data the user doesn't see | — | not rendered (and trivially editable — not secret) |
| `submit`, `reset`, `button`, `image` | buttons (prefer `<button>`, which can contain HTML) | — | — |

Rules of thumb:

- **`type="number"` only for numbers you'd do maths on.** A credit-card number, postcode or ID is a *string of digits*: use `type="text" inputmode="numeric"` (and maybe `pattern="[0-9]*"`). `number` drops leading zeros, accepts `e`, and scroll-wheels can change the value.
- The **display** format of a date follows the user's locale (their language-and-region setting — `09/28/2026` for a US locale), but the **value** is always ISO `2026-09-28` (the international standard date format, year-month-day).

### `inputmode` and `enterkeyhint`

`inputmode` chooses the virtual keyboard (the on-screen keyboard of a phone or tablet) **without** changing validation: `text`, `numeric`, `decimal`, `tel`, `email`, `url`, `search`, `none`. `enterkeyhint` labels the Enter key: `enter`, `done`, `go`, `next`, `previous`, `search`, `send`.

### Other common input attributes

`placeholder` (a hint, not a label), `value` (initial value), `required`, `readonly`, `disabled`, `autofocus` (use sparingly — it can disorient screen-reader users), `size`, `list` (points at a `<datalist>`), `multiple`, `accept`, `spellcheck`, `autocapitalize`.

### Self-check

1. Best markup for a 6-digit one-time code? → `type="text" inputmode="numeric" autocomplete="one-time-code"`.
2. What is the value of a date input showing 09/28/2026? → `"2026-09-28"`.
3. Why is `type="tel"` not validated? → Phone formats differ too much by country; add a `pattern` if you need one.

### TL;DR

Pick the most specific type for keyboards and validation; `number` only for real numbers; `inputmode` for keyboards without validation side effects.

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## 5.4 Choosing and grouping — `select`, `datalist`, `textarea`, `fieldset`, `output`, `progress`, `meter`

```
<!-- code/04_select_textarea_fieldset.html (excerpt) -->
<label>Country
  <select name="country" autocomplete="country">
    <option value="">-- choose --</option>
    <optgroup label="Europe"><option value="DE">Germany</option><option value="FR">France</option></optgroup>
    <optgroup label="Asia"><option value="IN" selected>India</option><option value="JP">Japan</option></optgroup>
  </select>
</label>

<select name="skills" multiple size="4"> … </select>

<input name="browser" list="browsers" value="Fire">
<datalist id="browsers"><option value="Chrome"><option value="Firefox"><option value="Safari"><option value="Edge">
</datalist>

<textarea name="message" rows="4" maxlength="200">Hello!
Line two keeps its line break.</textarea>

<fieldset>
  <legend>Delivery speed</legend>
  <label><input type="radio" name="speed" value="standard" checked> Standard (free)</label>
  <label><input type="radio" name="speed" value="express"> Express (+$5)</label>
</fieldset>
<fieldset disabled> <legend>Gift options (disabled fieldset)</legend> … </fieldset>

<progress max="100" value="70">70%</progress>
<meter min="0" max="100" low="60" high="85" optimum="20" value="91">91%</meter>
<input type="number" id="qty" name="qty" value="3"> x $12 = <output name="total" for="qty">$36</output>
```

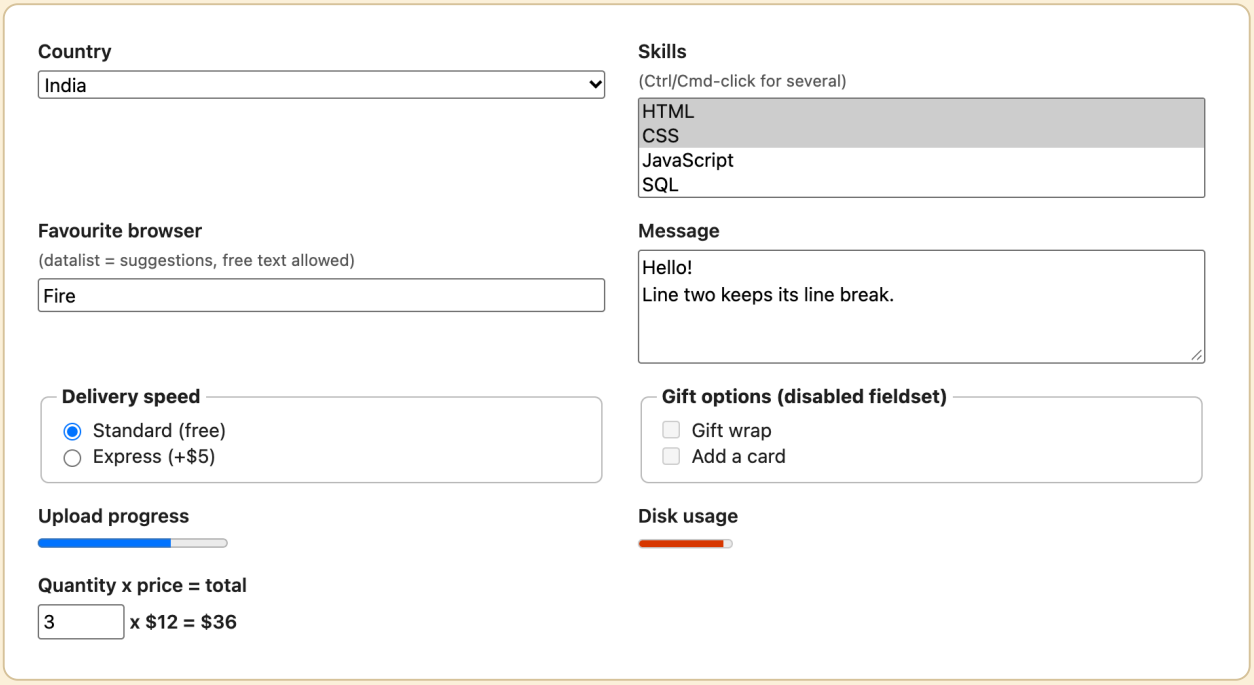


*Figure 5.4 — The disabled fieldset greys out and disables every control inside it; the meter turns red because 91 is above `high` and far from `optimum`.*

| Element | What it is | Key details |
|---|---|---|
| `<select>` | dropdown (a list that opens when clicked) of fixed options | Each `<option>` has a `value` (sent) and text (shown). A first option with `value=""` + `required` forces a real choice. `<optgroup label>` groups options. |
| `<select multiple>` | list box, several choices | Hard to use (Ctrl/Cmd-click); for more than a few options, a group of checkboxes is usually friendlier. |
| `<datalist>` | **suggestions** for an `<input list>` | The user may still type anything; it's autocomplete, not a restriction. |
| `<textarea>` | multi-line text | Initial value is its **content** (not a `value` attribute); `rows`/`cols`; `maxlength`; line breaks are preserved. CSS `resize` controls the drag handle. |
| `<fieldset>` + `<legend>` | groups related controls | Screen readers announce the legend with each control ("Delivery speed, Standard, radio button"). Use it for every radio group and checkbox group (the ARIA alternative is `role="radiogroup"`/`"group"` with `aria-labelledby`). `disabled` on the fieldset disables everything inside. |
| `<output>` | result of a calculation | `for` lists the ids of inputs it depends on; has an implicit live region (screen readers read changes aloud automatically), so updates can be announced. Not submitted. |
| `<progress>` | completion of a task | `value`/`max`; omit `value` for an indeterminate ("busy", no known percentage) bar. |
| `<meter>` | a measurement within a known range | `min`, `max`, `low`, `high`, `optimum` decide the colour (good/average/bad). For disk usage, ratings, password strength. Not for task progress. |

### Common mistakes

- Radio buttons without a `<fieldset>`/`<legend>` → the question ("Delivery speed") is never announced.
- Radios in one group with different `name`s → they don't exclude each other.
- `<select>` for 2–3 options (radios show all choices at once and need one click fewer) or for 200 countries without type-ahead support (jumping to an option by typing its first letters; a combobox — a text box combined with a filtered suggestion list — may be better).
- Using `<progress>` for a static measurement (use `<meter>`).

### Self-check

1. How do you make all controls in a section read-only-looking and unsubmittable at once? → `<fieldset disabled>`.
2. Datalist vs select? → Datalist suggests but allows free text; select restricts to its options.
3. Where does a textarea's initial value go? → Between its tags.

### TL;DR

`select` for fixed choices, `datalist` for suggestions, `textarea` for long text, `fieldset`+`legend` to group (always for radio groups), `output`/`progress`/`meter` for results and levels.

---

## 5.5 Built-in validation and the Constraint Validation API

### What it is

Browsers can validate forms without any JavaScript. You declare **constraints** with attributes; the browser checks them on submit, blocks submission if any fail, shows a message next to the first failing control, and exposes the result to CSS and JavaScript.

### The constraint attributes → the validity flag they set

| Attribute | Applies to | Fails when | validity flag |
|---|---|---|---|
| `required` | most inputs, select, textarea | empty / unchecked / nothing selected | `valueMissing` |
| `type="email"`, `type="url"` | input | value not shaped like an email/URL | `typeMismatch` |
| `pattern="regex"` | text-like inputs | value doesn't match the **whole** regex (regular expression — a mini-language for text patterns; it's implicitly anchored with `^…$`, i.e. must match from start to end) | `patternMismatch` |
| `minlength` / `maxlength` | text-like, textarea | shorter / longer — **only checked after the user edits the value** | `tooShort` / `tooLong` |
| `min` / `max` | number, range, date/time | below / above | `rangeUnderflow` / `rangeOverflow` |
| `step` | number, range, date/time | not a multiple of `step` from the base (`min`, else the initial `value`, else 0) | `stepMismatch` |
| (the browser) | number, date… | the user typed something unparsable | `badInput` |
| `setCustomValidity("msg")` | any | message is non-empty | `customError` |

Empty optional fields skip every check except `required` — an empty `pattern` field is valid.

Fields that are `disabled`, `readonly`, `type="hidden"`, or inside a `<datalist>` are **barred from constraint validation** (`willValidate === false`).

### Styling states: `:invalid` vs `:user-invalid`

`:invalid`/`:valid` match *immediately*, so a required field is red before the user has typed anything — a hostile experience. `:user-invalid`/`:user-valid` only match **after the user has interacted with the field or tried to submit**. They are Baseline 2023 (all major engines since November 2023; widely available since May 2026).

```
input:user-invalid { border-color: #c0392b; background: #fdeeee; }
input:user-valid   { border-color: #1e8449; }
```

### The JavaScript API

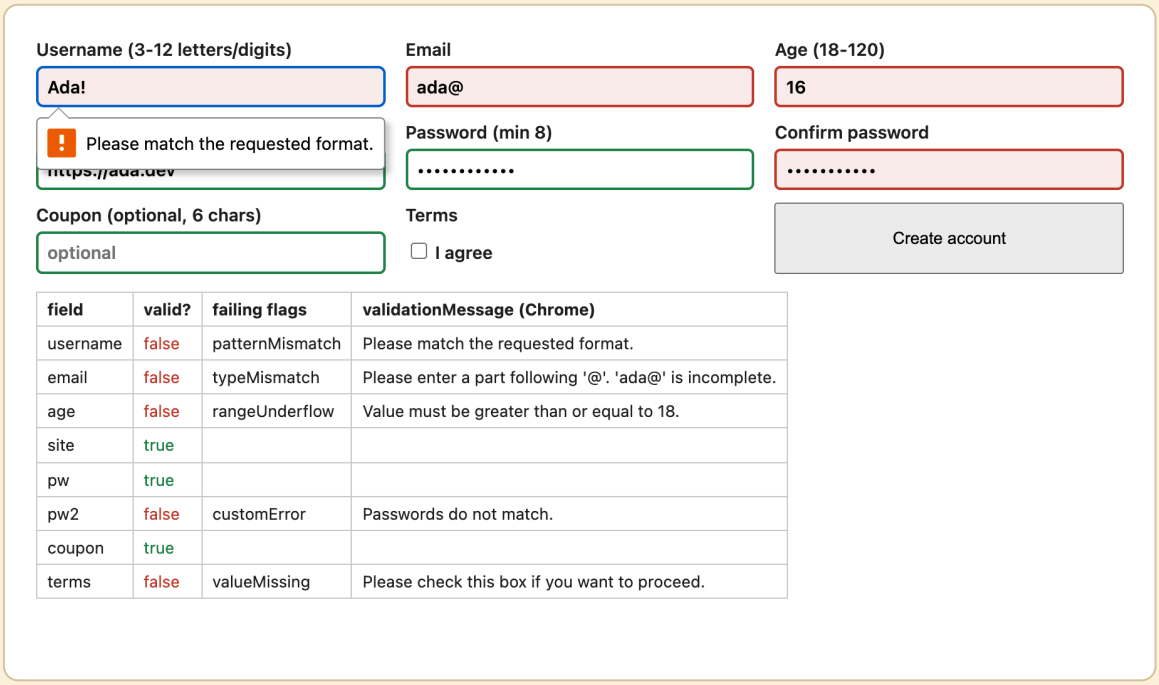
Each control has:

- `validity` — a `ValidityState` object with the flags above and `valid`.
- `validationMessage` — the browser's localised (translated into the user's language) message (empty when valid).
- `willValidate` — whether it takes part in validation at all.
- `checkValidity()` — returns true/false (a **boolean**); fires an `invalid` event on failure.
- `reportValidity()` — same, plus shows the browser's message bubble.
- `setCustomValidity(msg)` — add your own rule; **you must clear it with `''`** when the value becomes valid, or the field stays invalid forever.

Forms have `checkValidity()`, `reportValidity()` and `requestSubmit()`.

Demo in Chrome: a sign-up form with deliberately bad values; the script adds one custom rule (passwords must match), calls `form.requestSubmit()` (exactly like clicking the button), then prints each field's validity:

```
<!-- code/05_validation.html (excerpt) -->
<input name="username" required pattern="[a-z0-9]{3,12}" value="Ada!">
<input name="email" type="email" required value="ada@">
<input name="age" type="number" min="18" max="120" value="16">
<input name="site" type="url" value="https://ada.dev">
<input name="pw" type="password" required minlength="8" value="correcthorse">
<input name="pw2" type="password" required value="correcthors">
<input name="coupon" pattern=".{6}" placeholder="optional">
<input name="terms" type="checkbox" required>
<script>
  function checkMatch() {
    form.pw2.setCustomValidity(form.pw2.value === form.pw.value ? '' : 'Passwords do not match.');
  }
  form.pw.addEventListener('input', checkMatch);
  form.pw2.addEventListener('input', checkMatch);
  checkMatch();
  form.requestSubmit();
</script>
```




| field | valid? | failing flags | validationMessage (Chrome) |
|---|---|---|---|
| username | false | patternMismatch | Please match the requested format. |
| email | false | typeMismatch | Please enter a part following '@'. 'ada@' is incomplete. |
| age | false | rangeUnderflow | Value must be greater than or equal to 18. |
| site | true | | |
| pw | true | | |
| pw2 | false | customError | Passwords do not match. |
| coupon | true | | |
| terms | false | valueMissing | Please check this box if you want to proceed. |

*Figure 5.5 — Chrome blocked the submission, focused the first invalid field and showed its bubble; after the attempted submit, `:user-invalid` (red) and `:user-valid` (green) apply. The empty optional coupon is valid.*

The same API from Node, using jsdom:

```
// code/constraint_api.mjs (excerpt)
<input id="a" required>
<input id="b" type="email" value="not-an-email">
<input id="c" type="number" min="1" max="5" value="9">
<input id="d" pattern="[A-Z]{3}" value="abc">
<input id="e" pattern="[A-Z]{3}" value="">
<input id="f" required disabled>
<input id="g" minlength="5" value="ab">
…
d.setCustomValidity('That code is already taken.');
…
form.addEventListener('invalid', () => invalidEvents++, true);   // capture: invalid doesn't bubble
console.log('\nform.checkValidity() =', form.checkValidity(), '| invalid events fired:', invalidEvents);
```

Output:

```
a: willValidate=true valid=false valueMissing
b: willValidate=true valid=false typeMismatch
c: willValidate=true valid=false rangeOverflow
d: willValidate=true valid=false patternMismatch
e: willValidate=true valid=true -
f: willValidate=false valid=true -
g: willValidate=true valid=true -

after d.value = "ABC":           d: willValidate=true valid=true -
after setCustomValidity("..."):  d: willValidate=true valid=false customError | That code is already taken.
after setCustomValidity(""):     d: willValidate=true valid=true -

form.checkValidity() = false | invalid events fired: 3
```

Three details worth remembering from that output:

1. `e` — an empty optional field with a `pattern` is **valid**.
2. `f` — the disabled required field is barred (`willValidate=false`), so it never blocks submission. Chrome gives the same answer.
3. `g` — `minlength` is not enforced on a value set in HTML or by script; `tooShort` only appears after the user types.

## Trade-offs: built-in bubbles vs custom messages

| | Built-in UI | `novalidate` + your own messages (5.7) |
|---|---|---|
| Effort | zero | moderate |
| Message wording | generic, browser-controlled, localised to the browser language | your words, specific ("Postcode must be 5 digits") |
| Styling | bubble can't be styled | full control |
| Shows all errors at once | no — one at a time | yes (error summary) |
| Accessibility | acceptable; bubbles vanish quickly | best, if done as in 5.7 |

Most production design systems (a company's shared library of UI components, styles and rules) use `novalidate` and custom messages, **but still use the constraint attributes and API underneath** (`checkValidity()` still works with `novalidate`).

## Common mistakes

- Forgetting to reset `setCustomValidity('')` → the field can never become valid.
- `pattern` on `type="email"` duplicating what the type already checks, or patterns too strict for real names ("O'Brien", "José", single-letter surnames).
- Relying on client validation for security.
- Styling `:invalid` so forms load already red.

## In the interview

**30-second answer:** "HTML has declarative constraints — `required`, `type`, `pattern`, `min/max/step`, `minlength/maxlength` — which the browser checks on submit and exposes through `element.validity` flags, `validationMessage`, `checkValidity()` / `reportValidity()` and `setCustomValidity()` for custom rules. Style with `:user-invalid` so errors only show after interaction. Client-side validation is UX (user experience — it makes the form nicer to use); the server must re-validate."

- *Q: `checkValidity` vs `reportValidity`?* A: Both return a boolean and fire `invalid`; `reportValidity` also displays the message to the user.
- *Q: How do you validate "confirm password"?* A: On input, `setCustomValidity(match ? '' : 'Passwords do not match.')`.

## Self-check

1. Which flag does an out-of-range number set? → `rangeUnderflow` or `rangeOverflow`.

2. Why might a required field inside `<fieldset disabled>` not block submit? → Disabled controls are barred from validation.
3. `:invalid` vs `:user-invalid`? → The latter waits for interaction or a submit attempt.

### TL;DR

Declare constraints in HTML, read them with `validity`, add rules with `setCustomValidity` (and clear it), style with `:user-invalid`, and always validate again on the server.

---

## 5.6 `autocomplete` — help the browser fill the form

### What it is

The `autocomplete` attribute tells the browser and password managers **what a field means**, using standard tokens, so they can fill it correctly. It's also a WCAG 2.2 (Web Content Accessibility Guidelines, Chapter 3) requirement: success criterion 1.3.5 *Identify Input Purpose* (level AA) asks for it on fields collecting information about the user.

| Field | Token |
|---|---|
| Full name / given / family | `name`, `given-name`, `family-name` |
| Email, phone | `email`, `tel` |
| Username | `username` |
| Login password | `current-password` |
| Sign-up / change password | `new-password` (triggers strong-password suggestions) |
| SMS (text message) one-time code | `one-time-code` |
| Address | `street-address`, `address-line1`, `address-level2` (city), `postal-code`, `country` / `country-name` |
| Card | `cc-name`, `cc-number`, `cc-exp`, `cc-csc` (expiry date and the 3–4 digit security code) |
| Birthday | `bday` |
| Turn off (for things like a search of other people's names) | `off` |

Tokens can be prefixed with a section (`shipping street-address`, `billing postal-code`).

### Why it matters

Autofill turns a 2-minute checkout into seconds, reduces typos, and helps people with motor or memory disabilities. Password managers rely on `current-password` / `new-password` to tell sign-in from sign-up. WCAG 2.2 also added 3.3.7 *Redundant Entry* (don't make people re-type information they already gave in the same process) — autofill and "same as shipping" checkboxes help meet it.

### Common mistakes

- `autocomplete="off"` on login fields "for security" — it pushes users to weaker, reused passwords; browsers often ignore it for passwords anyway.
- Splitting phone numbers or dates into many boxes, which autofill can't fill.
- Blocking paste in password fields (hurts password managers; also runs against WCAG 2.2's 3.3.8 *Accessible Authentication*).

### TL;DR

Give every personal-data field its `autocomplete` token; `new-password` vs `current-password`; don't disable autofill or paste.

---

## 5.7 Accessible error messages

### The problem it solves

Built-in bubbles appear one at a time and disappear; colour alone ("the red one is wrong") fails colour-blind users (people who cannot tell certain colours, often red and green, apart); an error message far from its field is not announced when the field is focused. WCAG requires errors to be **identified in text** (3.3.1), fields to have **labels or instructions** (3.3.2), and fixes to be **suggested** when known (3.3.3).

### How it works — the pattern

Popularised by design systems such as the UK's GOV.UK Design System (the UK government's shared web components and patterns):

1. Add `novalidate` to the form so you control the messages; still use constraint attributes and `checkValidity()`.
2. On submit, check every field. For each failing one:
   - set `aria-invalid="true"` (screen readers say "invalid entry");
   - write a specific message into an element next to the field;
   - connect it with `aria-describedby="help-id error-id"` so it's read after the label when the field is focused.
3. Show an **error summary** at the top: a heading with the count and a list of links to each field (`href="#fieldId"`).
4. **Move focus to the summary** (it has `tabindex="-1"` so it can be focused by script) so screen readers announce it immediately.
5. Don't rely on colour: add an icon or the word "Error", and keep the red text at sufficient contrast.
6. Remove `aria-invalid` and the message as soon as the field is fixed.

```html
<!-- code/06_accessible_errors.html (excerpt) -->
<form id="f" novalidate>
  <div id="summary" tabindex="-1" hidden>
    <h2>There are 2 problems with your details</h2>
    <ul></ul>
  </div>
  <div class="field">
    <label for="email">Email address</label>
    <span class="help" id="email-help">We'll send the receipt here.</span>
    <input id="email" name="email" type="email" required autocomplete="email"
           aria-describedby="email-help email-err" value="ada@">
    <span class="error" id="email-err"></span>
  </div>
  <div class="field">
    <label for="zip">Postcode <span class="help">(required)</span></label>
    <input id="zip" name="zip" required autocomplete="postal-code" inputmode="numeric"
           pattern="[0-9]{5}" aria-describedby="zip-err" value="">
    <span class="error" id="zip-err"></span>
  </div>
  …
</form>
<script>
  const messages = {
    email: v => v.valueMissing ? 'Enter your email address.' : 'Enter an email address like name@example.com.',
    zip:   v => v.valueMissing ? 'Enter your postcode.' : 'Postcode must be 5 digits.',
    name:  v => 'Enter the name shown on your card.',
  };
  form.addEventListener('submit', e => {
    e.preventDefault();
    const problems = [];
    for (const input of form.querySelectorAll('input')) {
      const err = document.getElementById(input.id + '-err');
      if (input.checkValidity()) { input.removeAttribute('aria-invalid'); err.textContent = ''; }
      else {
        const msg = messages[input.name](input.validity);
        input.setAttribute('aria-invalid', 'true');
        err.textContent = msg;
        problems.push(`<li><a href="#${input.id}">${msg}</a></li>`);
      }
    }
    summary.hidden = problems.length === 0;
    summary.querySelector('ul').innerHTML = problems.join('');
    if (problems.length) summary.focus();
  });
</script>
```

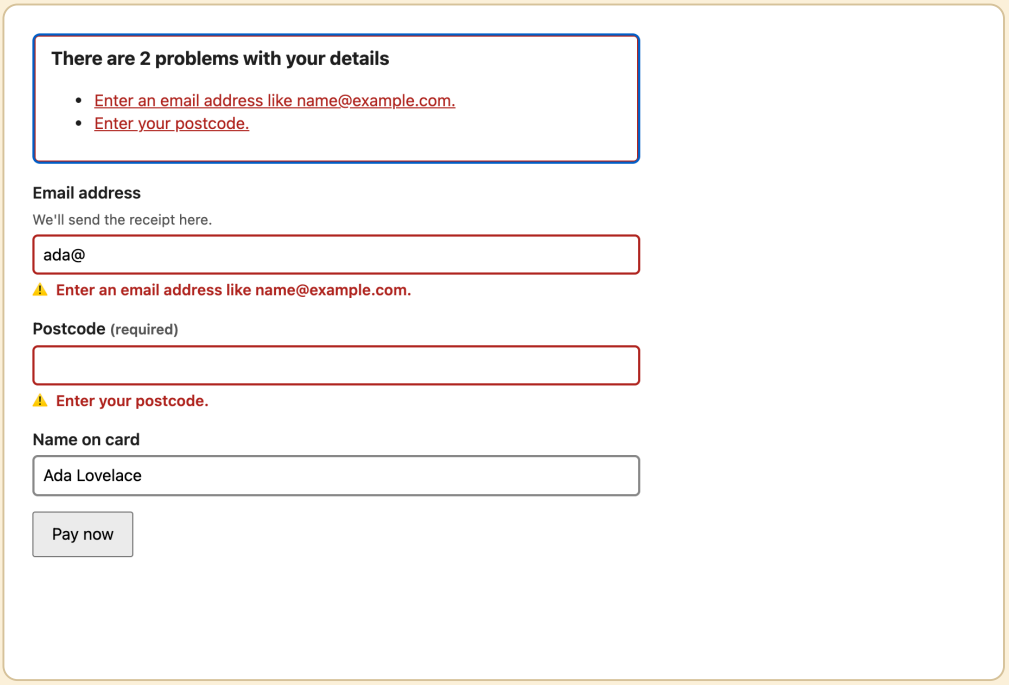



*Figure 5.6 — After one failed submit: summary focused (blue focus ring), each error has an icon + text next to its field, and messages say how to fix the problem.*

### Good error messages

- Say what's wrong **and how to fix it**: "Postcode must be 5 digits", not "Invalid input".
- Use the field's label words; no codes, no blame ("You entered an invalid…").
- Validate on submit (and optionally on `blur` — the event fired when a field loses focus — or, after the first submit, on `input`, fired on every change), not on every keystroke while the user is still typing.

### Common mistakes

- Error text not programmatically associated (linked in the markup so software knows they belong together; no `aria-describedby`).
- Only a red border.
- Clearing the whole form on error.
- Disabling the submit button until the form is valid — users can't discover *why* it's disabled. Let them submit and explain.

### In the interview

> **30-second answer:** "I use `novalidate` with the constraint API, then on submit mark each bad field with `aria-invalid`, show a specific text message next to it linked via `aria-describedby`, render an error summary with links at the top and move focus to it. Never colour alone, and keep the user's input."

### Self-check

1. Which attribute links a field to its error text? → `aria-describedby`.
2. Why `tabindex="-1"` on the summary? → So script can focus it without adding it to the Tab order.
3. What does `aria-invalid="true"` add? → Screen readers announce the field as invalid.

### TL;DR

Specific text messages next to fields (`aria-describedby`), `aria-invalid`, a focused error summary with links, never colour alone.

---

## Chapter interview round-up

1. **How do you make a data table accessible?** `<caption>`, `<th>` with `scope`, `thead/tbody/tfoot`, `headers` for irregular tables, scroll container for small screens.
2. **GET vs POST for forms?** GET = idempotent (repeating it has the same effect as doing it once — see System Design Fundamentals) reads, data in URL; POST = changes/private data in body; files need `multipart/form-data`.
3. **What isn't submitted?** Unnamed, disabled, unchecked controls, unselected options, buttons other than the one used, `<output>`.
4. **Why a `<label>` rather than a placeholder?** Accessible name, larger click area, persists while typing.
5. **How does built-in validation work and how do you customise it?** Constraint attributes → `validity` flags; `setCustomValidity`; `:user-invalid`; `novalidate` + custom UI for full control.
6. **What does `autocomplete="new-password"` do?** Tells password managers it's a new password (offer to generate/save), not a login.
7. **`<button>` inside a form refreshes the page — why?** Default `type="submit"`.

---

## Chapter TL;DR

- Tables are for data: `caption`, `thead`/`tbody`/`tfoot`, `th scope`, `headers` for complex cases; never for layout.
- A form sends **named, enabled, checked** controls to `action` via `method`; GET for reads, POST for changes, multipart for files.
- Every control needs a real `<label>`; group radios/checkboxes with `fieldset` + `legend`; buttons default to submit.
- Choose the most specific `type`; `inputmode` for keyboards; `number` only for numeric quantities.

- Built-in validation: `required`, `type`, `pattern`, `min/max/step`, `minlength/maxlength` → `validity`; `setCustomValidity` for extra rules; `:user-invalid` for styling.
- `autocomplete` tokens on personal-data fields; don't block paste or autofill.
- Accessible errors: `aria-invalid`, text messages via `aria-describedby`, focused error summary, never colour alone — and always validate on the server.

Next: Chapter 6 — CSS Fundamentals.